Name: Dheeraj Samtani
Roll No.: 2401098
Batch: T22
Subject: ASD&D

Experiment - 4

Experiment 4.1

Aim: To understand Continuous Integration, Jenkins installation.

LO mapped: LO3

Theory:

Continuous Integration is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

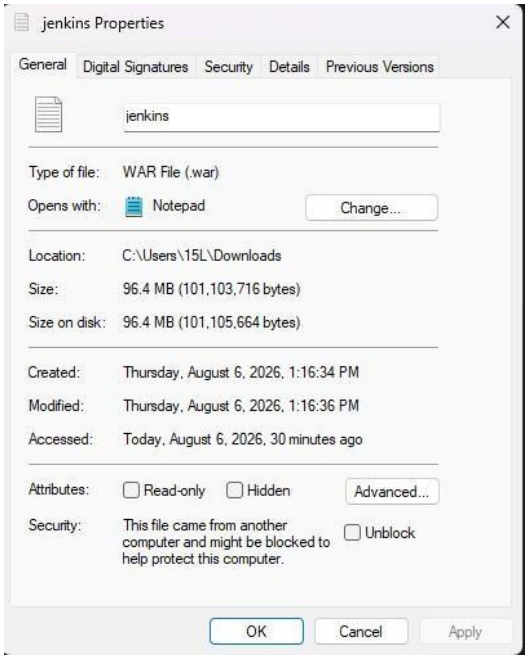
Jenkins is an open-source automation server that facilitates continuous integration and continuous delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supports a wide range of plugins and integrations, making it a versatile tool for automating various tasks in software development.

Steps to be followed while installing Jenkins:

I. **Java Installation:** Ensure Java Development Kit (JDK) is installed on your system.

```
C:\Users\15L>java --version
java 21.0.2 2024-01-16 LTS
Java(TM) SE Runtime Environment (build 21.0.2+13-LTS-58)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.2+13-LTS-58, mixed mode, sharing)
```

II. **Download Jenkins:** Get the latest Jenkins WAR file from the official website or package manager.



III. **Launch Terminal:** Open a terminal/command prompt on your system.

```
command: java -jar C:\Users\<User>\Downloads\jenkins.war --enable-future-java
```

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```
C:\Users\15L>cd Downloads

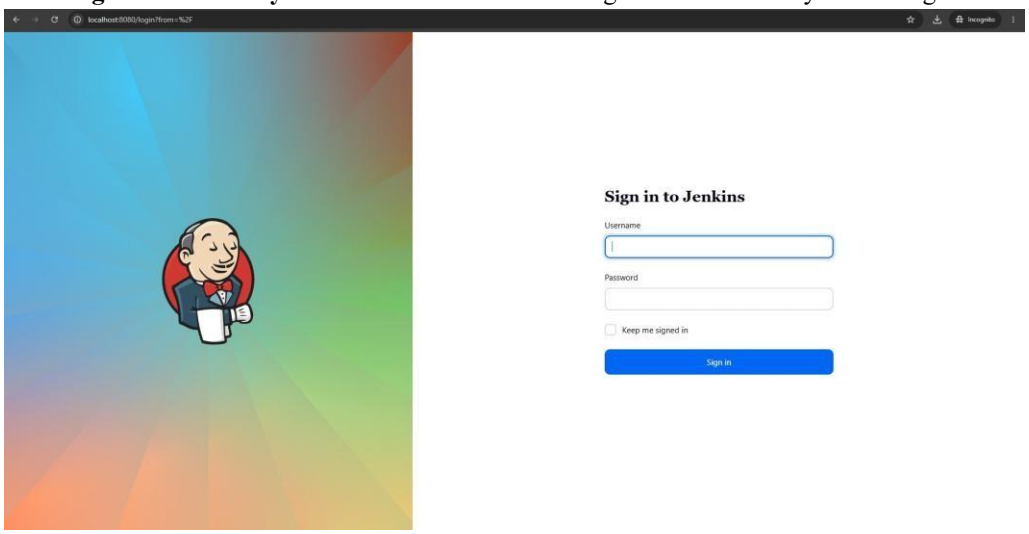
C:\Users\15L\Downloads>java -jar jenkins.war --enable-future-java
Running from: C:\Users\15L\Downloads\jenkins.war
webroot: C:\Users\15L\.jenkins\war
2026-08-06 07:51:24.191+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-06 07:51:24.273+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-06 07:51:24.334+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01
T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 21.0.2+13-LTS-58
2026-08-06 07:51:24.843+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /
, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-06 07:51:24.883+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-08-06 07:51:25.160+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\1
5L\.jenkins found at: $user.home/.jenkins
2026-08-06 07:51:25.238+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$Co
reContextHandler@5aa6202e{Jenkins v2.568.2,/,b=file:///C:/Users/15L/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$Co
reContextHandler$CoreToNestedHandler@3af9aa66{STARTED}}
2026-08-06 07:51:25.260+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@6d8
68997{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-06 07:51:25.261+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@2c383e33{ST
ARTING}[12.1.8,sto=0] @929ms
2026-08-06 07:51:25.261+0000 [id=61]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPor
t=disabled
2026-08-06 07:51:25.331+0000 [id=54]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
2026-08-06 07:51:25.385+0000 [id=66]    INFO    hudson.PluginManager#loadDetachedPlugins: Upgrading Jenkins. The last ru
nning version was 2.492.1. This Jenkins is version 2.568.2.
2026-08-06 07:51:25.387+0000 [id=68]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-06 07:51:25.600+0000 [id=66]    INFO    hudson.PluginManager#loadDetachedPlugins: Upgraded Jenkins from version
2.492.1 to version 2.568.2. Loaded detached plugins (and dependencies): [plugin-util-api.hpi, jackson3-api.hpi, gson-api
.hpi, display-url-api.hpi, scm-api.hpi, snakeyaml-engine-api.hpi, matrix-project.hpi, workflow-api.hpi, bootstrap5-api.h
```

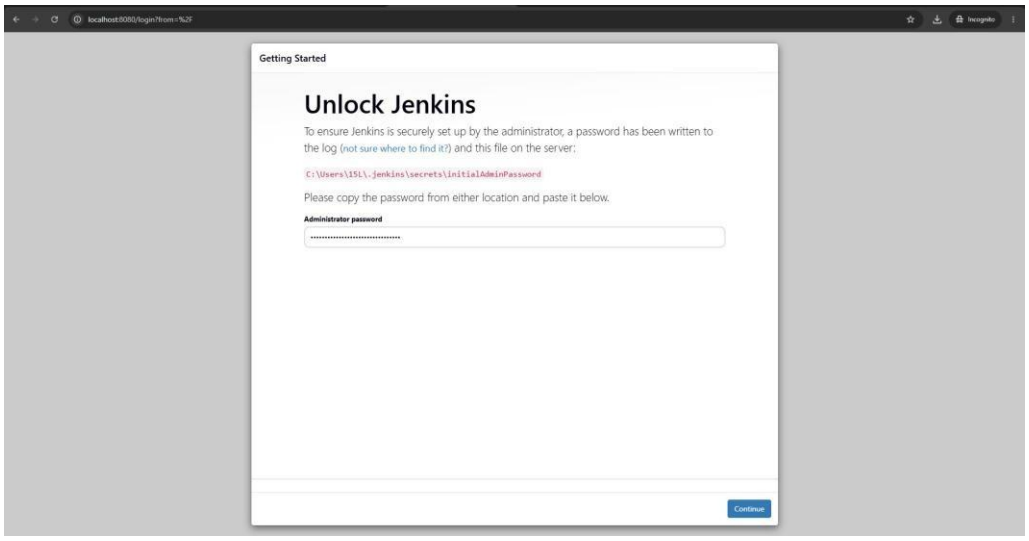
```
pi, jackson-annotations2-api.hpi, eddsa-api.hpi, commons-lang3-api.hpi, workflow-support.hpi, matrix-auth.hpi, workflow-
step-api.hpi, jquery3-api.hpi, caffeine-api.hpi, bouncycastle-api.hpi, prism-api.hpi, ionicons-api.hpi, asm-api.hpi, jax
b.hpi, checks-api.hpi, trilead-api.hpi, mailer.hpi, echarts-api.hpi, junit.hpi, font-awesome-api.hpi, script-security.hp
i, commons-text-api.hpi, instance-identity.hpi, structs.hpi, woodstox-core-api.hpi]
2026-08-06 07:51:27.263+0000 [id=78]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-06 07:51:27.710+0000 [id=92]    SEVERE  jenkins.InitReactorRunner$1#onTaskFailed: Failed Loading plugin Matrix A
uthorization Strategy Plugin v3.2.10 (matrix-auth)
java.io.IOException: Failed to load: Matrix Authorization Strategy Plugin (matrix-auth 3.2.10)
[CRLF]>  - Update required: Folders Plugin (cloudbees-folder 6.982.vf165a_16c9507) to be updated to 6.1026.ve06dfa_cf31c
3 or higher
        at hudson.PluginWrapper.resolvePluginDependencies(PluginWrapper.java:1018)
        at hudson.PluginManager$2$1.lambda$run$0(PluginManager.java:588)
        at org.jvnet.hudson.reactor.TaskGraphBuilder$TaskImpl.run(TaskGraphBuilder.java:175)
        at org.jvnet.hudson.reactor.Reactor.runTask(Reactor.java:304)
        at jenkins.model.Jenkins$5.runTask(Jenkins.java:1151)
        at org.jvnet.hudson.reactor.Reactor$2.run(Reactor.java:221)
        at org.jvnet.hudson.reactor.Reactor$Node.run(Reactor.java:120)
        at jenkins.security.ImpersonatingExecutorService.lambda$wrap$0(ImpersonatingExecutorService.java:66)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1144)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:642)
        at java.base/java.lang.Thread.run(Thread.java:1583)
2026-08-06 07:51:29.631+0000 [id=100]   INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-06 07:51:29.644+0000 [id=89]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-06 07:51:29.645+0000 [id=89]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-06 07:51:29.901+0000 [id=93]    INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-06 07:51:30.512+0000 [id=104]   INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-06 07:51:30.513+0000 [id=80]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-06 07:51:30.533+0000 [id=81]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-06 07:51:30.538+0000 [id=81]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updat
ed
```

```
2026-08-06 07:51:29.644+0000 [id=89]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-06 07:51:29.645+0000 [id=89]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-06 07:51:29.901+0000 [id=93]    INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-06 07:51:30.512+0000 [id=104]   INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-06 07:51:30.513+0000 [id=80]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-06 07:51:30.533+0000 [id=81]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-06 07:51:30.538+0000 [id=81]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updat
ed
2026-08-06 07:51:30.545+0000 [id=120]   INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates ser
ver
2026-08-06 07:51:30.545+0000 [id=70]    INFO    j.install.InstallState$Upgrade#applyForcedChanges: New API token system
configured with insecure options to keep legacy behavior
2026-08-06 07:51:30.546+0000 [id=86]    INFO    hudson.model.UserIdMapper#migrate: migrating C:\Users\15L\.jenkins\users
\users.xml
2026-08-06 07:51:30.559+0000 [id=86]    INFO    hudson.model.UserIdMapper#migrate: migrated C:\Users\15L\.jenkins\users\
admin_15206531508743068943 to C:\Users\15L\.jenkins\users\admin_5e58407f8dc291ba90b4d3f9e1a01b6aae2672753f7084bb03a5e2b4
5dc62c2f
2026-08-06 07:51:30.564+0000 [id=86]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-06 07:51:30.585+0000 [id=54]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-08-06 07:51:35.750+0000 [id=120]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file fo
r hudson.tasks.Maven.MavenInstaller
2026-08-06 07:51:37.404+0000 [id=120]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file fo
r hudson.tasks.Ant.AntInstaller
2026-08-06 07:51:40.028+0000 [id=120]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file fo
r hudson.plugins.gradle.GradleInstaller
2026-08-06 07:51:40.030+0000 [id=120]   INFO    hudson.util.Retrier#start: Performed the action check updates server suc
cessfully at the attempt #1
2026-08-06 07:52:13.466+0000 [id=150]   WARNING j.m.api.Metrics$HealthChecker#execute: Some health checks are reporting
as unhealthy: [plugins : There are 1 failed plugins: matrix-auth]
```

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IV. **Navigate to Directory:** Use the cd command to navigate to the directory containing the Jenkins WAR file.

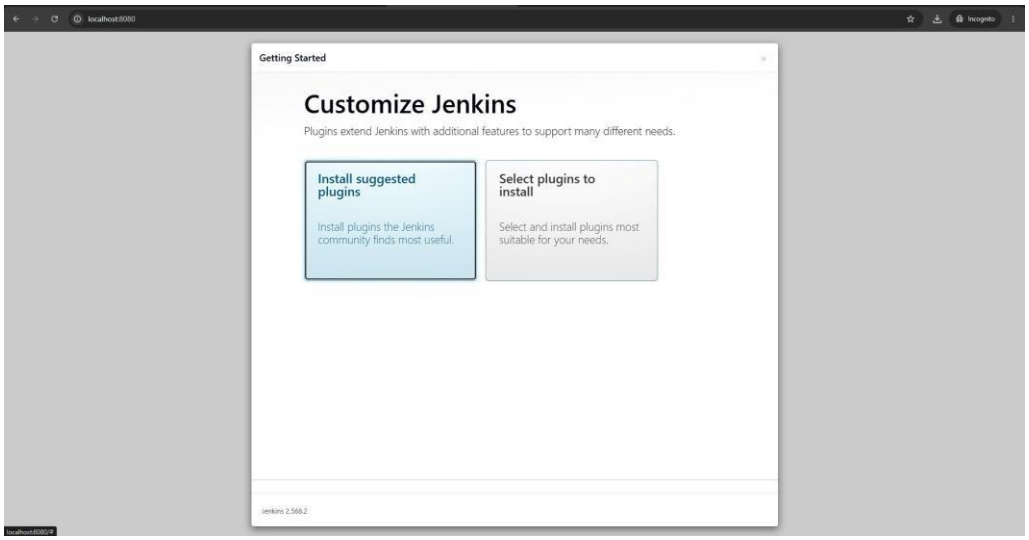




V. **Start Jenkins:** Run the command java -jar jenkins.war to start the Jenkins server.

VI. **Access Web UI:** Open a web browser and go to http://localhost:8080 to access the Jenkins setup wizard.

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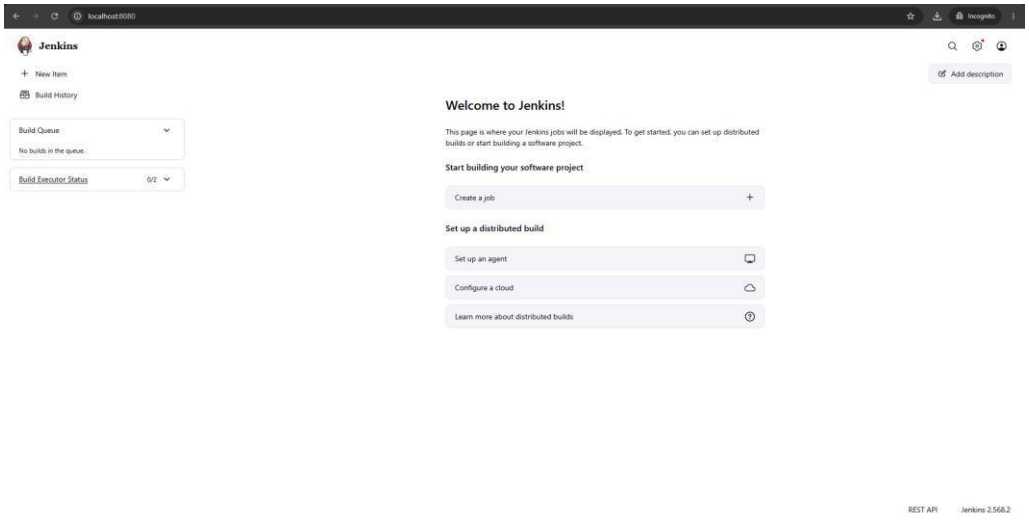
VII. **Unlock Jenkins:** Retrieve the initial admin password from the terminal and enter it in the setup wizard.

VIII. **Customize Plugins:** Choose recommended or custom plugins to install during setup.

IX. **Create Admin User:** Set up an admin user account with credentials.

X. **Jenkins URL:** Confirm the Jenkins URL and complete the setup process.

XI. **Dashboard:** Log in with the admin credentials to access the Jenkins dashboard.



Conclusion:

By this assignment we learned about Continuous Integration and the way to install Jenkins.

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Experiment 4.2

Aim: To build a pipeline using Jenkins.

LO mapped: LO3 **Theory:**

What is a Pipeline in Jenkins?

A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a continuous integration/continuous delivery (CI/CD) environment. In simple terms, it's a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key Benefits of Jenkins Pipeline:

- Automates CI/CD processes
- Provides better visibility into the build process
- Offers version control of build logic (Pipeline as Code)
- Supports complex workflows (parallel execution, conditional logic, etc.)

Types of Jenkins Pipelines:

There are two main types of syntax used to define Jenkins pipelines:

1. Declarative Pipeline

Simpler and more structured syntax. Designed for users who want an easy-to-read format. Introduced later to make pipelines more readable and maintainable.

Key Features:

- Uses pipeline {} block
- Organized into stages and steps
- Easier error handling
- Better validation in Jenkins UI

2. Scripted Pipeline

More flexible and powerful. Written in Groovy-based syntax. Gives full control over the build process. Suitable for advanced users.

Key Features:

- Uses node {} block
- Everything is written as code
- Allows complex scripting and logic
- Less user-friendly for beginners **1. Declarative Pipeline — Creation**

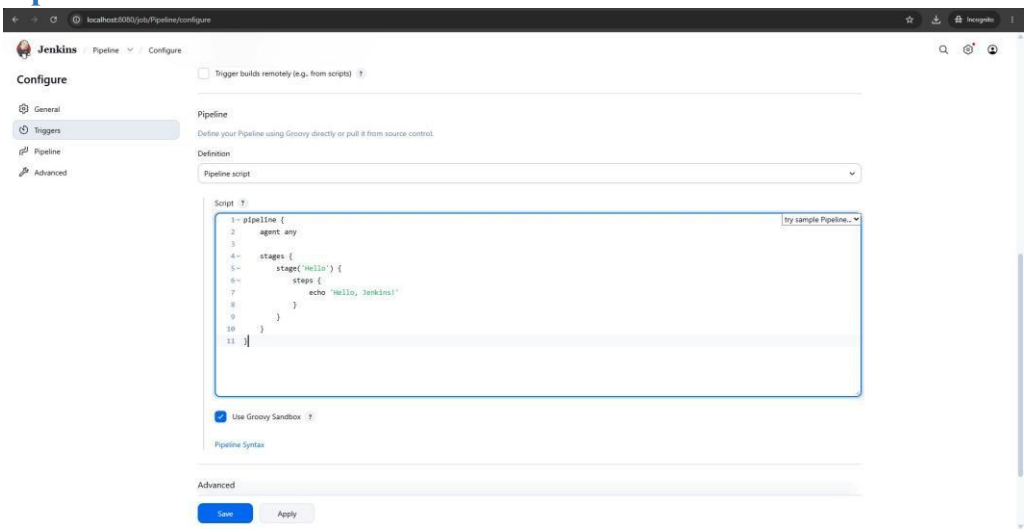
Code:

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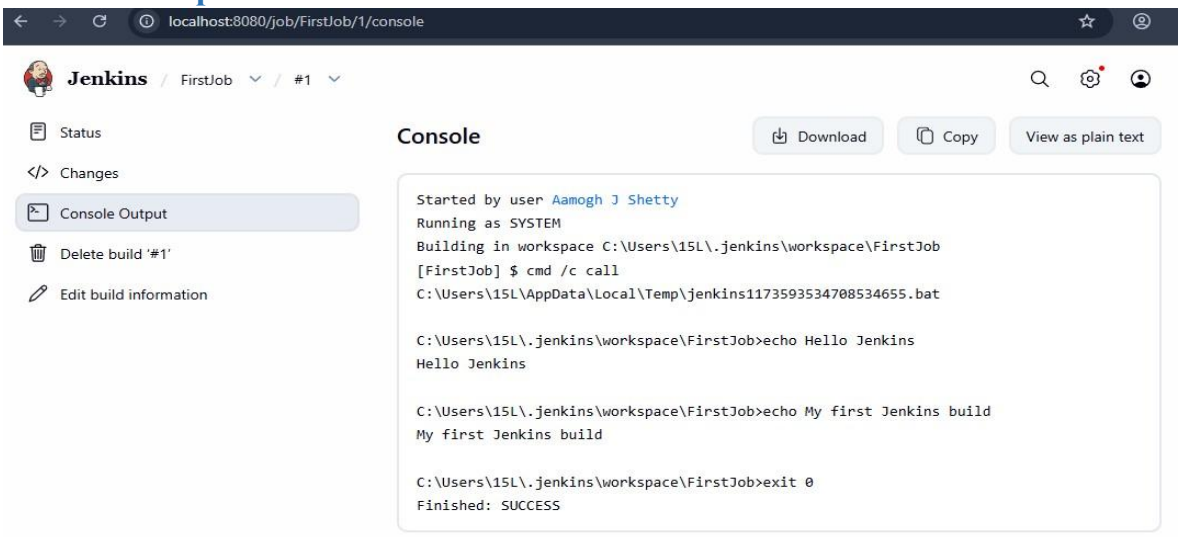
```
pipeline {
agent any

    stages {
        stage('Hello') {
steps {
                echo 'Hello, Jenkins!'
            }
        }
    }
}
```

Output:



2. Console output



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Conclusion: LO1, LO3

In this assignment, we learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.